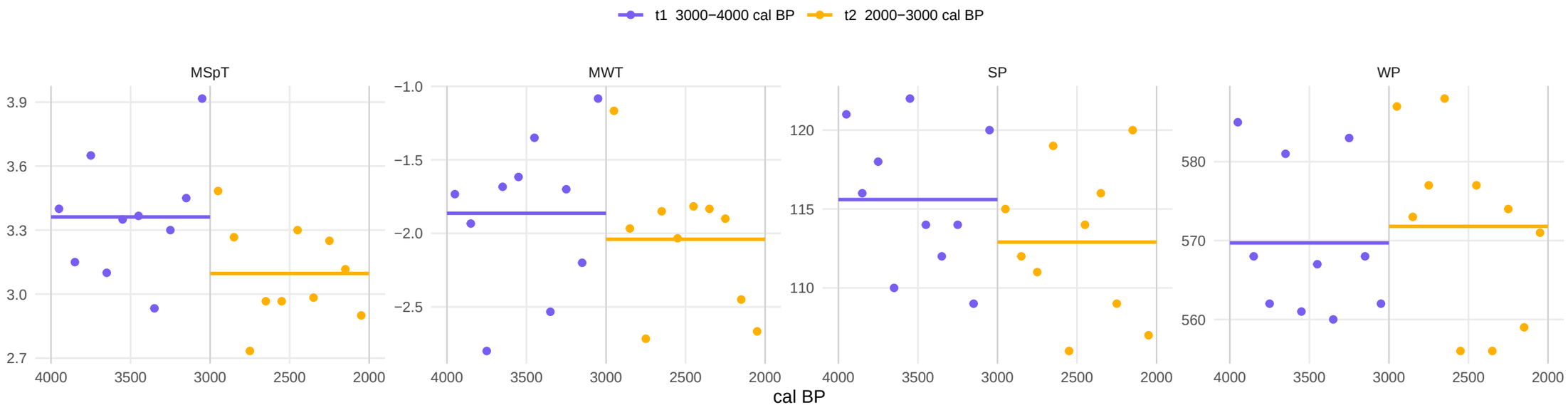
How the currently plotted metrics are built − one real interval at PantherPotholes (NOCA), bin offset 0

Every number is taken from R/stages/09_tracking_metrics.R, not recomputed. Bins are 1000 yr wide; 100 Bacon age draws; this is 1 of the 10 bin offsets.

A. A bin's CLIMATE is the mean of its 10 native 100−yr CHELSA slices

Points = the native slices. Horizontal bar = the bin mean. The bin is a window on the TIME axis and TraCE21k slice ages are known calendar dates, so this side of the calculation is FIXED − no age draw can move it.

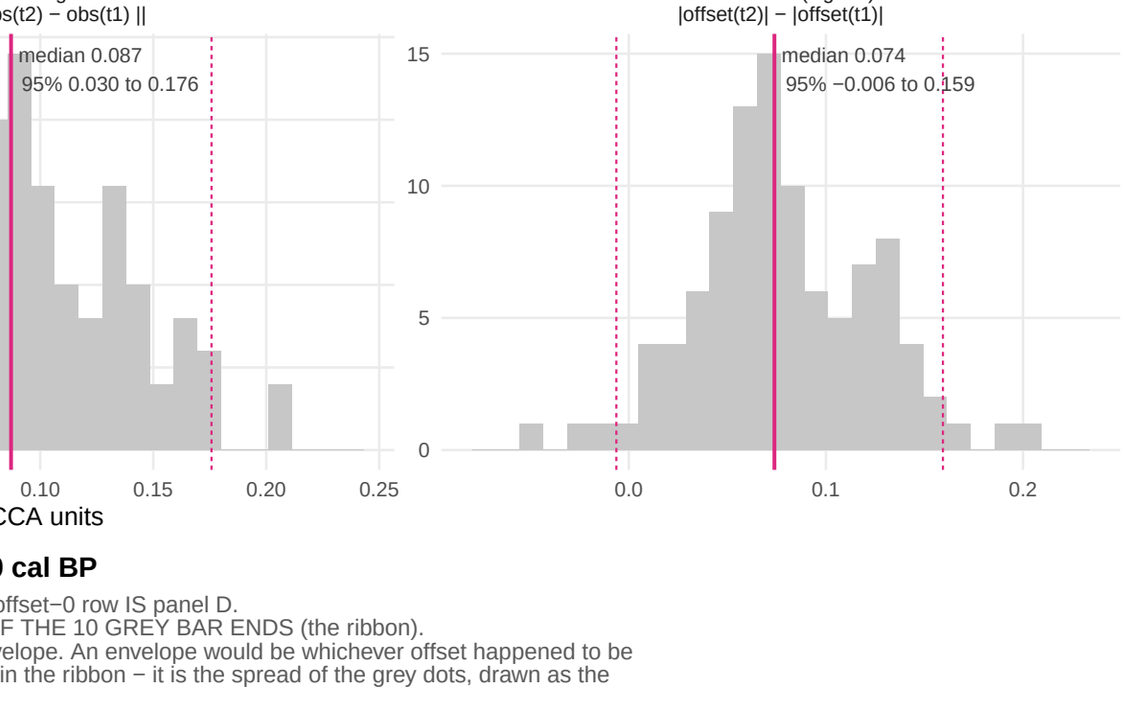


B. ...but which SAMPLES fall in it is not

One line per macrofossil sample near the 3000 cal BP edge: its 95% Bacon age range, dot = median age. 25 of 85 straddle the edge, so which bin they belong to changes from draw to draw.

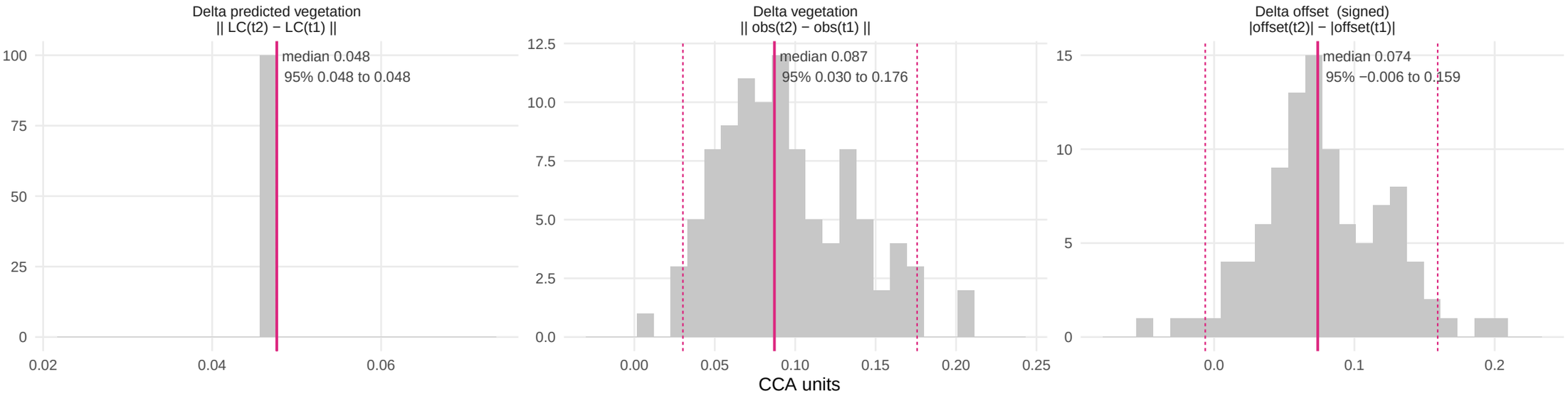
C. Each bin gets TWO positions in the ordination

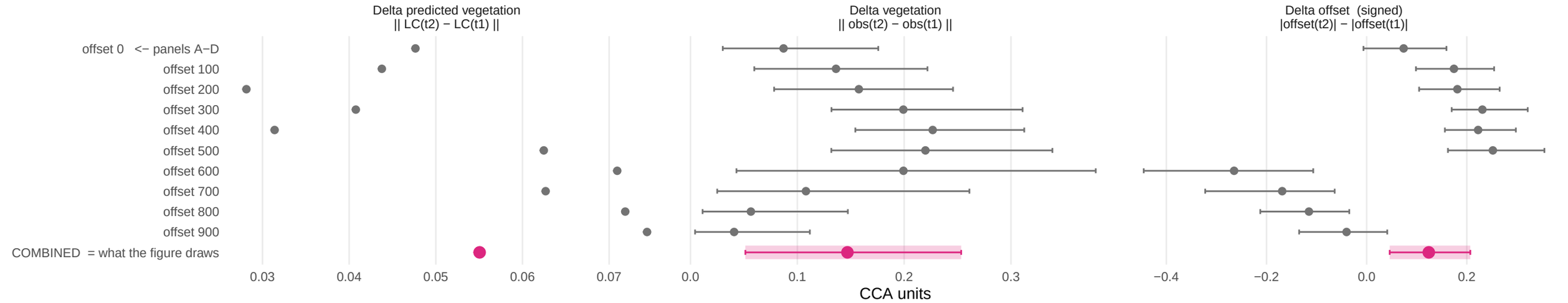
Triangle = pred, from the bin's climate (one point − fixed). Faint dots = obs, one per age draw; large dot = their median. The dashed line is offset = pred − obs. The metrics are the lengths of these things and how they change from t1 to t2.



D. The currently plotted metrics for this interval, over 100 age draws

LC−predicted vegetation displacement is a SPIKE because panel A is fixed. The other two spread, because panel B is not. The pink median and dashed 95% are the realization−level quantities summarized in the authoritative tracking tables for every interval, both sites, and 10 bin offsets.



E. ...and how the 10 bin offsets become ONE point and ONE band, at t = 3000 cal BP

Grey rows: each offset ON ITS OWN − dot = median over its 100 age draws, bar = its 2.5−97.5. The offset−0 row IS panel D.
Pink row: dot = MEDIAN OF THE 10 GREY DOTS (the retained nested estimate); band = MEDIAN OF THE 10 GREY BAR ENDS (the ribbon).
The ribbon is therefore the TYPICAL age−uncertainty band − not a sum of the offsets, and not an envelope. An envelope would be whichever offset happened to be widest, so it would keep growing the more offsets were tried; the median does not. Bin phase is NOT in the ribbon − it is the spread of the grey dots, drawn as the thin lines on the final figure.